江苏省苏州中学 国际部
Suzhou High School – International Division

SZZX-ID
BP Physics

Magnetic effect of a current

Class Outline

Title of Unit	Introduction to magnetic field
Title of Lesson	Magnetic effect of a current
Starter Activity	Video: magnetic field around a current carrying wire and a solenoid
Direct Instructions	Teaching Strategies: Demonstration, Lecture, Collaboration, Questioning, Inquiry
	- demonstration: magnetic field of a solenoid - Lecture: right hand grip rule - Inquiry: worksheet to apply the right-hand grip rule
	Differentiation: Content, Instructional Strategies, the Classroom, Products, Teacher
	- Pair work on the worksheet - Explain the worksheet
	International Mindedness:
Lesson Objectives	By the end of the lesson, students will be able to: • Use the right hand grip rule to determine the direction of magnetic field around a straight wire and a solenoid • Understand and apply notations into and out of the page plane
ATLs	Identify unit ATLs and how they are being addressed in the lesson
	Thinking: Applying key ideas and facts in new contexts – apply understanding of E field in B field
IB Learner Profile	Link the lesson to the characteristics of the IB Learner attribute
	Inquirer, thinker
Teaching Activities	- Demonstration: magnetic field in and near a solenoid - worksheet: draw magnetic field lines (with help form simulator); right-hand rule - Video: earth's magnetic field
Plenary Session	Checking for understanding / Homework / Q & A / Reflection etc.
	Video: why does the Earth have a magnetic field

Vocabulary

Solenoid/coil 线圈

Concentric circles 同心圆

Curl 弯曲

Magnet field: a Long, Straight Wire

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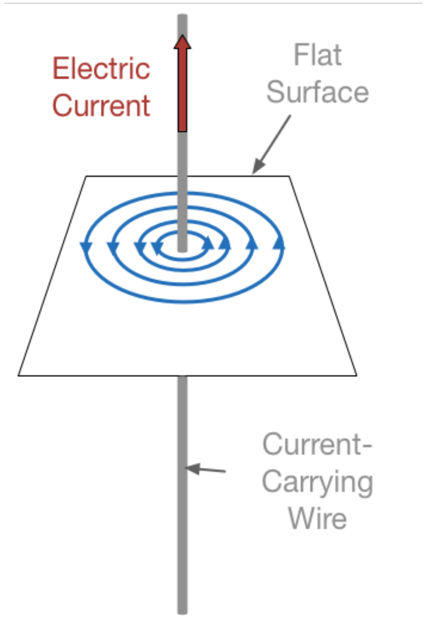
Any current-carrying conductor will produce a magnetic field around it.

The magnetic field due to a long, straight wire features:

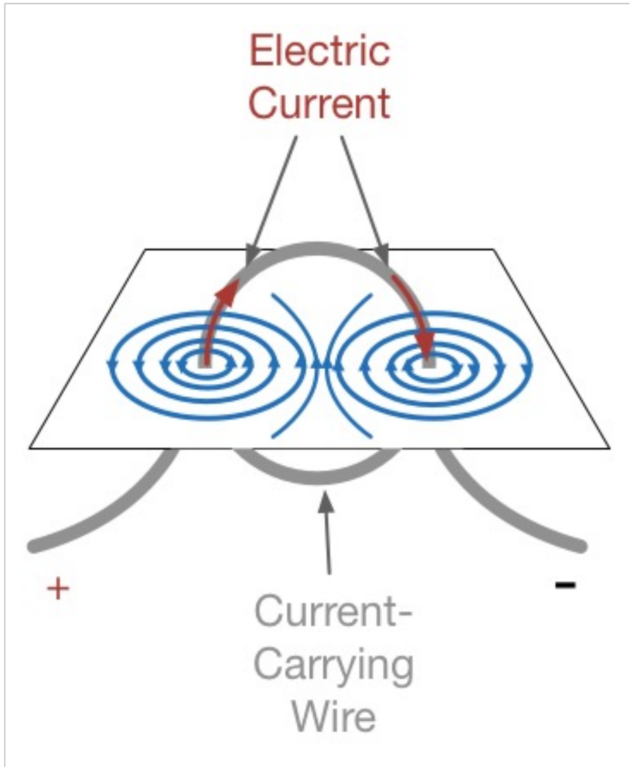
- **Concentric circles;**
- **Magnetic field strength decreases as radius increases**
- **The larger the current, the stronger the field**



Magnetic field - Flat, Circular Coil

The magnetic field due to a flat, circular coil is shown.

- **Concentric circles due to each wire**
- **Fields point in the same direction in the center, due to different directions of current**
- **Magnetic field strengthens in the center of the coil.**



Magnetic field – Solenoid

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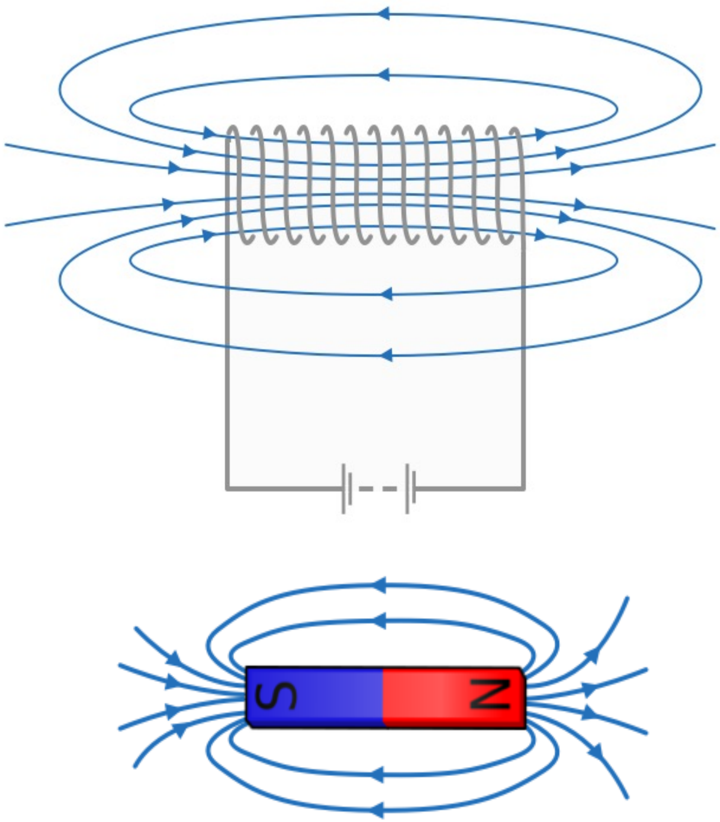

A solenoid is a series of linked flat, circular coils. The magnetic field due to a solenoid is shown.

- Magnetic field has a similar pattern to a bar magnet;
- A uniform field is found in the center

Right Hand Grip Rule

The direction of the magnetic field around a wire can be found using the "right-hand grip rule"

- **Use right hand**
- **The thumb points in the direction of conventional current (if electrons are charge carriers, direction of current is the opposite direction of electron flow)**
- **The fingers curl around in the direction of the magnetic field.**

Right Hand Grip Rule

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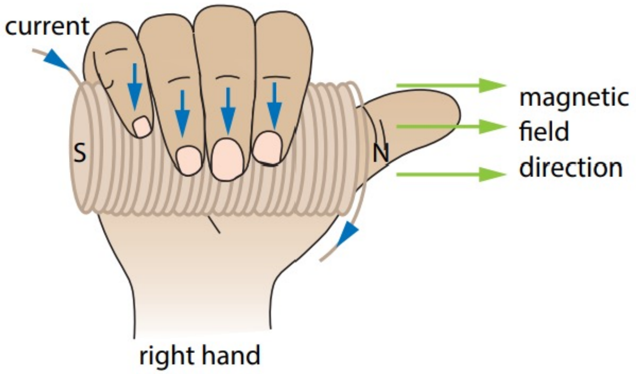

The direction of the magnetic field around a solenoid can be found using the "right-hand grip rule", too

- **Use right hand**
- **The fingers curl in the direction of the current**
- **The thumb points in the direction of the magnetic field**

Magnetic Field Strength, B

The magnetic field strength, B, can be thought of as the density of magnetic field lines per square meter (also called Magnetic Flux Density, B).

It is a vector.

It has units of Tesla, T.